Cascading Multimodal Feature Enhanced Contrast Learning for Music Recommendation

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***Abstract*—Representation learning remains one of the most important but challenging tasks within industrial music recommendation systems. In the context of the Matthew effect, item exposure frequency demonstrates substantial inequality, leading to the Harry Potter problem for popular items and the long-tail issue for less interacted items, collectively impairing the adequacy and accuracy of representation learning. In this paper, to alleviate the negative impact of bias on representation learning in music recommendation systems, we propose a unified model based on introducing the unbiased Cascading Multimodal Feature, called CMF4Rec. Specifically, with our cascading feature enhancement module, we implement a dual-stage representation enhancement strategy. In the first stage, the pivotal subsequence is extracted from the coarse-grained similarity sequence derived from cascading multimodal features, which is subsequently aggregated to generate the enhanced representation of the candidate item. Moreover, in the feature interaction module, the enhanced representation is crossed with user behaviors to capture the diverse and dynamic interests of users. Furthermore, we employ contrastive learning and design an auxiliary contrastive task to provide high-quality gradients for the main recommendation task. We demonstrate the effectiveness of this model with extensive experiments on public and industrial datasets. Moreover, the deployment of CMF4Rec in a real music recommendation system has also yielded significant improvements.**

***Index Terms*—music recommendation, representation learning, multimodal feature, contrastive learning, data bias**

I. Introduction

With the exponential expansion of the Internet [1], the issue of information overload has become pervasive in the realm of music, making it difficult for users to access tracks that align with their personal preferences. In response to this, music recommendation systems have been introduced to curate and prioritize information according to the specific needs and preferences of individual users. Besides, by improving the efficiency of information dissemination, platforms stand to gain substantial benefits. As one of the most important prerequisite for precise prediction of user behaviors, sufficient learning and accurate modeling of representations are difficult problems for music recommendation systems.

Despite methods based on deep learning like [2], [3] and [4] achieve significant success, representation learning typically exhibits the problem of inadequate learning caused by data bias. In particular, items in the music recommendation system can be divided into head items and tail items according to the frequency of interaction due to the existence of bias such as exposure bias and popularity bias [5], [6]. For head items, the high frequency of interaction could make connections between unrelated items, thus introducing inaccurate gradients, which is known as the **Harry Potter problem** [7]. While for long-tail items, the scarcity of user feedback impedes the acquisition of adequate and effective gradients, resulting in inadequate representation learning, known as the **long-tail problem**.

To achieve comprehensive learning of representations, the mainstream approach concentrates on feature interaction and enhancement. For example, [8] decouples item processes and user distributions, introducing an adapter to softly shift training attention to tail items. [9] designs a framework comprising three self-supervised signals to produce high-quality feature representations. Despite the encouraging performance, the bias in user-item interaction data used in these models has not been mitigated, which means that the gradients learned in the training process are still suffering from inaccurate problem.

Therefore, to achieve adequate and effective gradient acquisition, instead of only use interaction data, we also pay attention to the inherent information of items termed as **cascading multimodal features**, which can be divided into two categories:

- **Basic features.** This kind of features refers to the attributes possessed by most items and is easily accessed, such as artists and genres of songs.
- **Multimodal features.** Considering the problem of sparse features of long-tail items, we further obtain multimodal features, such as the audios and images of songs.

It is worth noting that relying solely on the single item is not sufficient and solid, we also introduce a similar sequence determined by the cascading multimodal features to enrich the feature space for recommendation.

In this paper, we propose a unified model named CMF4Rec based on the unbiased **C**ascading **M**ultimodal **F**eature for music **Rec**ommendation systems to alleviate the negative impact of bias on representation learning. CMF4Rec contains two primary components: the cascading feature enhancement module and the feature interaction module. Specifically, the first module adopts a dual-stage feature enhancement strategy. Initially, based on the cascading multimodal features, we utilize the offline Approximate Nearest Neighbor (ANN)

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algorithm to create a coarse-grained similar sequence. Subsequently, an attention-based network is designed to generate a more relevant subsequence and aggregate it as an enhanced representation for the candidate item. In the feature interaction module, the enhanced candidate representation is utilized as a new feature input, which plays a pivotal role in capturing the diverse and dynamic interests of the users by crossing with user behavior sequences. In addition, we employ Contrastive Learning (CL) and design an auxiliary comparison task to minimize the distance between the candidate item and its enhanced representation, to generate high quality gradients. In summary, the contributions of our proposed model CMF4Rec are as follows:

- We design our approach for addressing the constraints of relying solely on biased interactive data in representation learning. To tackle this inadequacy, we propose to introduce the cascading multimodal features to provide unbiased information and high quality gradients.
- We propose a unified model named CMF4Rec, which contains two key modules to provide additional information on capturing user interests and an auxiliary task to supply meaningful gradient signals.
- Extensive offline experiments on both public and industry datasets verify the effectiveness of our CMF4Rec. Online A/B test also shows significant improvements.

II. Related Work

A. Representation Learning

Representation Learning is a long-standing challenge in recommendation systems: the existence of bias brings difficulties to the learning of representations for both head and tail items.

To address this challenge, several strategies to refine the loss function have been proposed. For example, [10] defines an ideal loss functions, based on which a method is developed to predict items with the highest correlation rather than the highest click probability. [11] introduces a theoretical framework to measure data overlap by associating with each sample with a small neighboring region and further design a re-weighting scheme to re-balance the loss. Another common method is resampling. For example, [12] proposes a strategy based on high variance to avoid false negative instances by selecting candidates with high variance.

For deep learning-based approaches, they enhance the accuracy of recommendation results by effectively capturing complex and high-order feature interactions and lead to remarkable result. For example, [13] learns fine-grained feature interactions via bilinear function and dynamically learns the importance of features via the squeeze-excitation network mechanism. [14] is a framework that applies feature corruption and recovery on multi-field categorical data to learn capable representations of features and instances. In order to ensure the efficiency and accuracy of the model, [15] is an adaptive feature interactive learning model, which learns the feature interaction in complex vector space by space mapping according to Euler's formula. However, these models still face the problem of inadequate representation learning caused by incomplete attributes of items.

B. Self-Supervised Learning

Self-Supervised Contrastive Learning (SSCL) [16] aims to learn representations by using an auxiliary task to increase the similarity score for positive samples and decrease for negative samples. It has achieved significant success in fields such as computer vision (CV) and natural language processing (NLP) [17], [18]. Recently, contrastive learning is widely applied in recommendation systems [19], [20]. For example, [21] leverages contrastive learning obtains self-supervised signals from the inherent correlations in the data, and enhances data representations through pre-training methods. [9] consists of three components, i.e., contrastive loss, feature alignment, and field uniformity to directly generate high-quality feature representations. [22] is designed to address the issue of data sparsity by introducing two auxiliary matching tasks, which enhance the relationship between users and items, thereby improving accuracy. [23] presents a flexible framework based on multimodal features, utilizing contrastive learning to infuse recommendation domain knowledge into the model.

III. Method

A. Problem statement

Base Model. In this paper, we take the ranking model as an example. With $\mathcal{P}$ and $\mathcal{Q}$ being collections of users and items, ranking model aims to predict the probability that the user i engages with the item j within the given context, denoted as $y_{i,j} \in (0,1)$, where $i \in \mathcal{P}$ and $j \in \mathcal{Q}$. Each sample is composed of input of the instance $X = (X_p, X_q, X_C)$ and output of a binary label y indicates click or not. X_p, X_q and X_C represents the feature set associated with user, item, and context, respectively. Denoting the ranking model as F, the prediction task can be formulated as:

$$P(y = 1|X) = F(X). \tag{1}$$

Cascading Multimodal Features. In the proposed CMF4Rec, X_q representing items encompasses not only basic features such as ID and category, but also incorporates cascading multimodal features, which contains both basic features and auxiliary features with different modalities. For the basic, like numerical features as artists and genres of song, they have relatively low acquisition costs and are typically included in the data infrastructure of most industrial recommendation systems. However, the acquisition of multimodal features requires the utilization of specialized algorithms for embedding transformation. For instance, leveraging Natural Language Processing (NLP) algorithms to extract feature from textual content as vectors is essential. In general, we design a feature embedding layer to translate these numerical and multimodal cascading multimodal features into embeddings.

Framework. There are two key modules in the proposed CMF4Rec: Cascading Feature Enhancement Module and Feature Interaction Module, which are shown in Figure 1. The first module employs a dual-stage approach to enhance candidate item representations. In the second module, user interests are fused from the raw and the enhanced representations as a novel feature in the prevailing recommendation framework. In the rest of this section, we will detail the two modules.

Similar Sequence. Considering that relying solely on the single candidate item is not sufficient and solid, a cascading multimodal features-based similar sequence is computed for each item to provide more gradients. Specifically, we employ the offline Approximate Nearest Neighbor (ANN) algorithm, such as Faiss, to perform similarity retrieval to generate a similar sequence for each item, and items in the sequence are listed in descending order of similarity. It should be noted that this similar sequence represents the initial output of the dual-stage cascading feature enhancement module within our CMF4Rec, which constitutes a critical component of the entire model. Given its pivotal role, we provide a thorough explanation here.

B. Cascading Feature Enhancement Module(CFEM)

1) Item Feature Encoder: In the first stage, the coarse-grained similar sequence is obtained through concatenation of cascading multimodal features and then the most similar items are obtained through a search for similarity in the vector space. Although efficient, it unavoidably introduces noise. This item feature encoder is used to

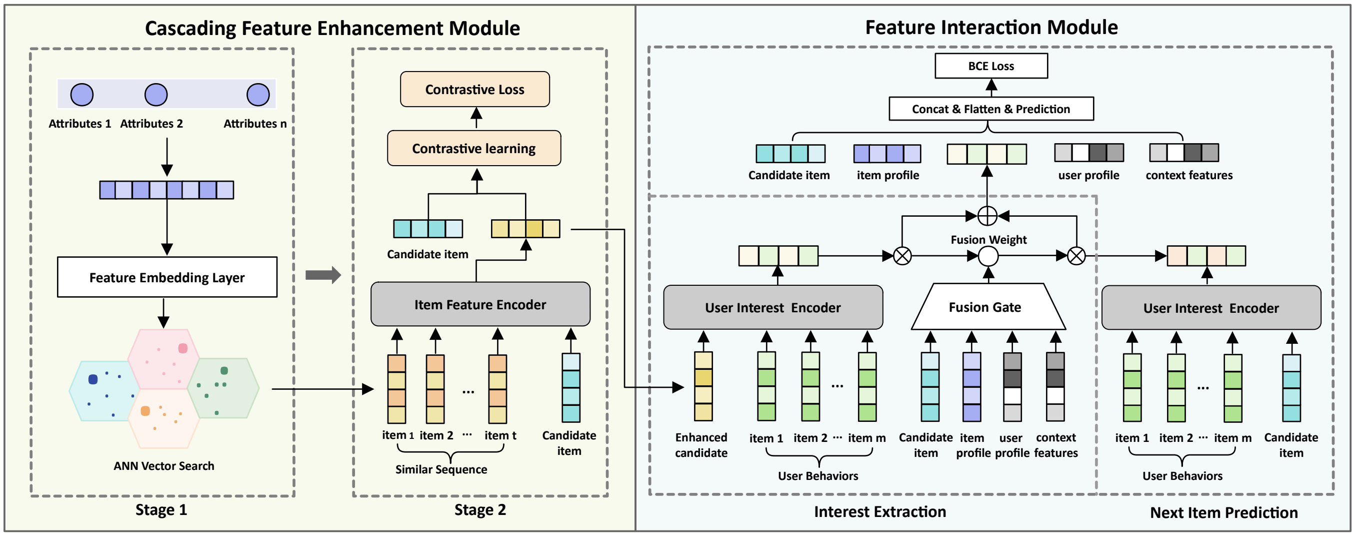

Fig. 1. The framework of the CMF4Rec. The dual-stage cascading feature enhancement module shown in the left part is designed to first generate a similar sequence for each candidate item. In the second stage, the most relevant items in the similar sequence are extracted, and an auxiliary contrastive task is designed to generate unbiased gradients. The right section shows the structure of the feature interaction module, in which we firstly capture user interest on the basis of original and enhanced candidate item. Secondly, we generate the final input feature for the next item prediction task in recommendation systems.

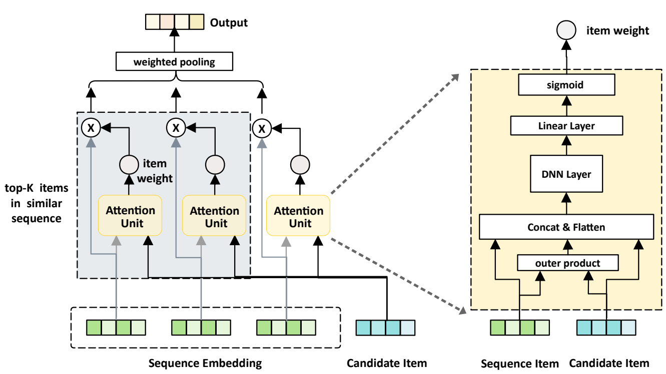

Fig. 2. The framework of the item feature/user interset encoder. Although these two encoders share the same structure, the parameters are different.

extract key information in the similar sequence and further reduce the influence of noise on representation learning.

Specifically, this encoder is a shared attention-based structure with a sequence and an individual item as inputs. The structure is shown in Figure 2. The attention mechanism is used to calculate how each item in the sequence relates to the input item.

In the CFEM, the inputs are embedding of the candidate item e_j and its corresponding similar sequence represented as $E_s^j = [e_1; ..; e_t] \in \mathbb{R}^{t\times d}$, where t is the length of this sequence and d is the embedding size. To infuse relevance information into the similar sequence, we initially combine the raw sequence with trainable positional representations $P \in \mathbb{R}^{t\times d}$. Formally, similar sequence with position embeddings E_p^j for candidate item e_j is defined as:

$$E_p^j = [e_1^p; e_2^p; ...; e_t^p] = [e_1 + p_1; e_2 + p_2; ...; e_t + p_t]. \tag{2}$$

To identify the items in the coarse-grained similar sequence that exhibit the highest relevance with the candidate item, the attention mechanism is employed to quantify the correlation between the sequence items and the candidate item. In particular, we firstly calculate attention scores W_a through a feed-forward network with attention mechanism $\alpha(\cdot)$ as Eq. 3.

$$W_a = \sum_{n=1}^{t} \alpha(e_j, e_n). \tag{3}$$

Secondly, we select top-k items with the highest scores, considering them to be strongly associated with the candidate item in terms of both cascading multimodal features and item representations, thereby providing comprehensive and reliable additional information for the candidate item. Besides, these k items in E_p^j are chosen to form the representation of similar sequence h_p^j based on weight W_a.

$$h_p^j = f(E_p^j, W_a, k) = \sum_{n=1}^{k} w_n e_n^p. \tag{4}$$

As an aggregated sequence representation, h_p^j significantly enriches the feature space of candidate items, which can be regarded as the enhanced representation of the candidate item e_j. For example, for popular items, h_p^j enables the model to discern the attributes that contribute to their popularity. In contrast, for tail items with sparse features, h_p^j supplements the information, thereby enhancing their representation.

2) Contrastive Task: Considering that the enhanced candidate representation h_p^j calculated above consists of the key information in the similar sequence that is highly correlated with the candidate item e_j, inspired by the success of SSL, we take these two representations as a pair of training positive samples.

In conventional contrastive models [24], each e_j often takes the other h_p^j in the same batch as negative pairs, and InfoNCE [25], [26] is utilized to minimize the distance between positive samples and maximize it between negative samples accordingly. However, constructing and calculating negative pairs often results in a significant escalation of computational expenses, particularly in terms of inference time, which is deemed unacceptable in practical recommendation systems.

Consequently, we chose an improved approach using only positive samples for loss computation by h_p^j and e_j, thereby circumventing the costs of negative sample computation, enhancing the practicality and efficiency of the system. With the batch size B, we utilize the l_2 distance $\|\cdot\|_2^2$ to calculate the loss of the contrast learning module:

$$\mathcal{L}_{CL} = \frac{1}{B}\sum_{n=1}^{B} \|h_p^{jn} - e_j^n\|_2^2. \tag{5}$$

C. Feature Interaction Module (FIM)

In CMF4Rec, the feature interaction module undertakes the crucial tasks of user interest analysis and candidate item scoring.

1) Interest Extraction: In addition to the similarity sequence of the candidate item, we also introduce the behavior sequence E_s^i of the i-th user to capture his/her preference. Considering that h_p^j extracted from CFEM contains the intrinsic attribute properties of the candidate item, it can enrich the feature space of the candidate item, which makes sense for enhancing the representation learning capability of our model. User interests are calculated from two aspects: the original candidate item e_j and the enhanced candidate representation h_p^j.

To assign varying weights to different historical behaviors when deriving user interest representations, thus reverse-activating users' historical interests based on the current candidate item, we use the user interest encoder depicted in Figure 2, which has the same structure with the item feature encoder while have independent parameters:

$$h_c^i = f'(e_j, E_s^i) = \sum_{n=1}^{m} \alpha(e_j, e_n) = \sum_{n=1}^{m} w_n e_n, \tag{6}$$

$$h_s^i = f'(h_p^j, E_s^i) = \sum_{n=1}^{m} \alpha(h_p^j, e_n) = \sum_{n=1}^{m} w'_n e_n, \tag{7}$$

where f' is the user interest encoder, m is the behavior sequence length , w_n and w'_n are the weights for the n-th item in behavior sequence E_s^i.

h_c^i and h_s^i respectively aggregate user interest from the perspectives of original candidate item and enhanced candidate representation based on cascading multimodal features. To dynamically and adaptively inject supplementary information to the final user preference representation, a network utilizing gating mechanism is designed to calculate their weights. In particular, we first concatenate the e_j with other features, such as its attribute features e_a^j, which consist of cascading multimodal features of this candidate item and context information as bias, to serve as the input to weight calculation. It can be formulated as follows:

$$\gamma = \sigma(MLP(e_j \oplus e_a^j \oplus u_a^i \oplus e_c)), \tag{8}$$

where σ is the sigmoid activation function and $\oplus$ is the vector concatenation operation. u_a^i represents the user's attribute, and e_c is the contextual features. Then, user's interest augmented with supplementary information can be calculated as:

$$h_i^j = \gamma h_c^i + (1-\gamma) h_s^i. \tag{9}$$

2) Next Item Prediction: In the prevailing recommendation model, user preference h_c^i and other features, including user profile u_a^i, item attributes e_a^j and context features e_c are often concatenated to form the input representation.

The difference between our model and existing methods is that, the feature space of the candidate items not only encompasses the basic attributes but also incorporates the cascading multimodal features. Additionally, in our model, h_i^j derived from both original and enhanced candidate item is used to replace h_c^i. The final input to the next item prediction task can be formulated as:

$$h_{fin}^j = h_i^j \oplus e_j \oplus e_a^j \oplus u_a^i \oplus e_c. \tag{10}$$

MLP module is often utilized as the final prediction layer with h_{fin}^j as its input, in which the output normalized by the sigmoid activation function to obtain the predicted result $\hat{y}$. Afterward, the widely-used negative log-likelihood loss is employed as the primary loss function:

$$\mathcal{L}_{Rec} = -\frac{1}{N}\sum_{i=1}^{N}(y_i \log(\hat{y_i}) + (1-y_i)\log(1-\hat{y_i})). \tag{11}$$

We add this main loss and contrastive loss together to supervise the model training. The total loss is:

$$\mathcal{L}_{Totle} = \lambda \mathcal{L}_{CL} + \lambda_{rec}\mathcal{L}_{Rec}, \tag{12}$$

where the λ_{rec} and λ are the weight coefficients.

IV. Experiments

A. Experimental setup

1) Datasets: We conducted extensive experiments on both an industry and two public datasets. The statistics of the three datasets are presented in Table I. Considering that ranking models require labels represented by 0 and 1 to construct positive and negative samples, we selected two public datasets: Spotify[1] [27] and 30Music[2] [28]. The Spotify dataset contains user music listening sequences with associated behavioral labels. The target behavior is defined as the "end play" label. In this dataset, audio features such as danceability and beat strength are exploited as cascading multimodal features. For 30Music, it contains the playlists of users and the "love" tag is recorded. We use the artist and genre information as cascading multimodal features. For Spotify and 30Music, according to previous research [29], we sorted all user interactions in chronological order and divided them into training and testing sets in an 8:2 ratio. For the industrial dataset, we collect user logs from the music recommendation scenarios during one week. We use the data of the first five days to train our model and the data of the last two days for testing. We not only take advantage of the basic features, but also incorporate multimodal information such as album covers and audio features into our CMF4Rec.

TABLE I
Statistics of experimental dataset.

Dataset	#User	#Item	#Samples	#Positive
Spotify	0.16M	3.7M	1.2M	3.4%
30Music	0.02M	4.5M	16M	9.4%
Industrial	7.0M	6.0M	21.8M	3.2%

2) Baselines: We compare our CMF4Rec with three types of classic recommendation methods. The first (FM [30], DeepFM [3], Wide&Deep [2]) focuses on second- and higher-order feature interactions, the second (DIN [4], DIEN [31]) adds behavior sequence on top of features, and the third (CL4CTR [9]) integrates auxiliary tasks into the model.

For the first type of baselines, we process user behavior sequences by pooling and input them as features.

3) Evaluation Metrics: To evaluate the performance of all methods, we employ widely-used metrics, AUC (Area Under ROC) and Logloss (cross entropy). It is worth noting that even a minor improvement in AUC or reduction in Logloss at the 0.001-level holds significance for ranking tasks [32].

[1] http://research.spotify.com/datasets/music-streaming-sessions

[2] https://recsys.deib.polimi.it/datasets/

TABLE II

OVERALL PERFORMANCE COMPARISON. THE BEST AND SECOND-BEST RESULTS ARE BOLD AND UNDERLINED, RESPECTIVELY.

	Spotify		30Music		Industrial	
	AUC	Logloss	AUC	Logloss	AUC	Logloss
FM	0.7590	0.5971	<u>0.8857</u>	<u>0.1927</u>	0.7165	0.2774
DeepFM	0.7607	0.5944	0.8767	0.2011	0.7184	0.2764
Wide&Deep	0.7648	0.5919	0.8771	0.2177	0.7302	0.2746
DIN	0.7622	0.5849	0.8843	0.1944	<u>0.7364</u>	<u>0.2722</u>
DIEN	0.7617	0.5850	0.8810	0.1964	0.7218	0.2767
CL4CTR	**0.7765**	<u>0.5257</u>	0.8830	0.2094	0.7118	0.2785
CMF4Rec	<u>0.7727</u>	**0.4571**	**0.9056**	**0.1830**	**0.7498**	**0.2678**
%Improve	-0.49	13.07	2.25	5.30	1.80	1.62

4) Implementation Details: We implement CMF4Rec with Tensorflow and train the model using a 40G A100 GPU. For the industrial dataset, the embedding size is set at 100, the learning rate is set to be 0.0001 and the batch size is set to be 1536, the loss weights $\lambda = 0.005$. For the Spotify and 30Music datasets, the embedding size, the learning rate and the batch size are set to be 20, 0.001 and 1024, respectively. And for all datasets, we set $k = 60$, $t = 20$ and use Adam as the optimizer for all datasets. Each experiment is repeated 3 times, and the average results are reported.

B. Performance Evaluation

Table IV-B presents the experimental results of CMF4Rec and baselines. There are three conclusions can be drawn: (1) Our model achieved the best performance across all datasets, which can be attributed to the incorporation of cascading multimodal features and the use of similarity sequence, which serves as supplement for candidate items and provide more unbiased gradients to our model. (2) For models like FM, even with the addition of similar sequence through pooling, the effectiveness falls short compared to methods like DIN. This underscores the necessity of fully utilizing sequence information. On the 30Music dataset, FM emerged as the second-best model. We attribute this to the dataset's limited feature types, where feature interactions may sufficiently capture user interests. On the 30Music dataset, FM emerged as the second-best model. We attribute this to the dataset's limited feature types, where feature interactions may sufficiently capture user interests. (3) On large industrial dataset, CL4CTR performs worse than DIN on one hand, which we attribute to the full benefit of user interaction sequences. On the other hand, the performance of CL4CTR is also inferior to the FM. We believe that the possible reason is that the features of this dataset are more complex, and a large gap in the same feature field is existed. The field uniformity may cause the loss of key information.

C. Ablation Study

TABLE III

ABLATION STUDY OF CMF4REC.

Dataset	Metric	w/o Feature	w/o SSL	w/o Bias	CMF4Rec
Spotify	AUC	0.7644	<u>0.7693</u>	0.7680	**0.7716**
	Logloss	0.5917	<u>0.5699</u>	0.5711	**0.5673**
30Music	AUC	0.8642	<u>0.9017</u>	0.8990	**0.9055**
	Logloss	0.2160	<u>0.1871</u>	0.1872	**0.1830**
Industrial	AUC	0.7328	0.7446	<u>0.7490</u>	**0.7507**
	Logloss	0.2725	0.2692	<u>0.2678</u>	**0.2672**

We introduce three variants of our model: **w/o Feature:** removes the introduction of cascading multimodal features. **w/o SSL:** eliminates the contrastive learning module, and instead utilizes the encoder directly to obtain similar sequence representations. **w/o Bias:** The similar sequence of each item is calculated by approach of collaborative filtering based on the biased interactive data. The performance of these three variants on all datasets is shown in table III. which proves the effectiveness of each module. Specifically, we have the following three observations.

Firstly, among these three variants, the performance of the w/o Feature variant is the worst across all datasets, effectively demonstrating that the introduction of cascading multimodal features can provide valuable supplementary information for the items, greatly facilitating the comprehensive learning of item representations by the model.

Secondly, in the w/o SSL variant, both AUC and LogLoss are not only worse than the CMF4Rec but also underperform the w/o Bias variant. This indicates a gap between the introduced similarity sequence and the candidate items, highlighting the necessity of using contrastive learning to align these two representations and provide high-quality gradients. However, the performance of this variant is still better than that of the w/o Feature variant, suggesting that even in the presence of the gap, the introduction of additional information can still yield benefits.

Thirdly, in the w/o Bias variant, the performance is better than the previous two variants, indicating that the introduction of additional information can bring supplementary gradients to the model. However, compared with the CMF4Rec, the correlation between items still exists in the popular items in the similar sequence calculated by using biased interactive data, leading to suboptimal results.

D. Representation Learning Analysis

To validate the enhancement of CMF4Rec on representation learning, we bucket the test set of the industrial dataset based on occurrence frequency. The bucket ID decreases with decreasing occurrence frequency. We compute the corresponding Logloss between Wide&Deep and DIN, and represented it using a bar graph. The line ΔLogloss represents the improvement over the DIN approach after applying our model. The experimental results are shown in Figure 3.

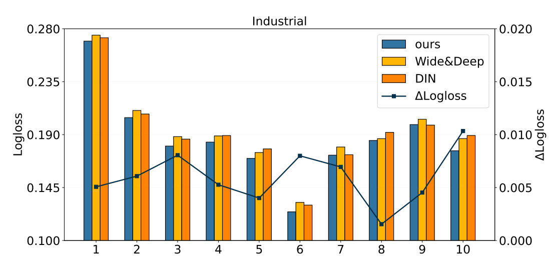

Fig. 3. The improvement of different buckets between CMF4Rec, Wide&Deep and DIN.

The results in Figure 3 clearly demonstrate that our CMF4Rec exhibits noticeable improvement in loss for both popular and long-tail items. This suggests that leveraging objective and unbiased cascading multimodal features mitigates the bias resulting from skewed data distribution to a certain extent. Consequently, it provides a richer source of higher-quality gradients for representation learning.

E. A/B Test

We conduct an A/B test in the industrial online recommendation system in five core scenarios to measure the benefits of CMF4Rec compare with the last version of our online-serving model. The results are shown in the Table IV.

As shown in the table above, our proposed CMF4Rec significantly improves performance in all scenarios, demonstrating the effectiveness of the proposed method.

TABLE IV
ONLINE RESULTS OF CMF4REC. S_i IS THE i-TH SCENARIO

Model	S_1	S_2	S_3	S_4	S_5
CMF4Rec	+3.30%	+1.5%	+3.20%	+0.80%	+1.64%

V. CONCLUSION

In this paper, we propose a music recommendation model CMF4Rec for the purpose of mitigating the effect of biased data and tackle the representation learning challenge. The proposed method leveraging the cascading multimodal features, which include base attribute features and multimodal features for providing unbiased information. What's more, we further calculate a similar sequence for each item on the basic of cascading multimodal features to enrich the feature space. Additionally, an auxiliary comparison task is designed to generate high quality gradients for the next item prediction task. Offline and online experiments validations demonstrate the effectiveness of the introduction of unbiased cascading multimodal features in addressing the representation learning problem.

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